JOHANSSON ENGINEERING

Supervision Program Manual — Pressurised Vessels

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Author	N. Johansson

1. Purpose

This manual defines **how to create, implement, and maintain an ongoing supervision program** for pressurised vessels. It is the complete methodology reference for the Vessel Supervision System (JDS-PRJ-MEC-002).

A supervision program is the documented system that ensures every pressurised vessel under your responsibility is regularly monitored between formal inspections. Swedish regulation AFS 2017:3 (Chapter 2, Section 3) requires every operator to have such a program. This manual tells you how to build one.

> **In plain language:** This manual teaches you to build a supervision program from scratch, run it day-to-day, and improve it year after year.

2. Scope

This manual covers: - Pressure vessels (tanks, receivers, heat exchangers, autoclaves) - Safety accessories (safety valves, rupture discs, pressure switches) - Pressurised piping systems where included in the regulatory scope

It does **not** cover: - Formal recurring inspections by accredited bodies (see JDS-PRO-004) - Equipment design, manufacture, or commissioning (see PED 2014/68/EU) - Equipment repair or modification procedures

3. Definitions

TERM	MEANING
Supervision program	A documented system defining what checks are performed, how often, by whom, and how results are recorded
Supervision round	A single execution of scheduled checks — a walk-around, instrument reading, or functional test
Ongoing supervision	The continuous responsibility of the operator to ensure safe operation between formal inspections
Formal inspection	Periodic inspection by an accredited body (Class A) or competent person (Class B) — external, internal, or pressure test
Operator	The person or organisation responsible for the equipment in service
Competent person	A person with documented training and experience to perform the assigned supervision tasks

4. Regulatory Foundation

4.1 Base Regulation

AFS 2017:3 — Use and Inspection of Pressurised Equipment — is the primary Swedish regulation. Chapter 2 establishes that the operator must ensure ongoing supervision is performed and documented.

4.2 Amendments (Consolidated)

AMENDMENT	KEY CHANGES
AFS 2019:1	Clarified supervision scope for piping systems, refined competence provisions
AFS 2020:10	Extended inspection deadlines during extraordinary circumstances, accepted digital documentation and remote verification
AFS 2022:2	Updated classification thresholds, revised competence requirements for supervision personnel

4.3 Key Regulatory Requirements

The regulation requires the operator to:

1. **Have a supervision program** — documented, site-specific, covering all equipment in scope
2. **Define check intervals** — based on equipment type, operating conditions, and risk
3. **Assign competent personnel** — with documented qualifications
4. **Document every round** — records must be retained for the life of the equipment
5. **Act on findings** — deviations must trigger corrective action
6. **Review the program** — at least annually, update as needed

> For the full regulatory summary and paragraph references, see JDS-RPT-MEC-003.

5. Building a Supervision Program

5.1 Overview

Building a supervision program follows five steps:

```
SURVEY → ASSESS → DESIGN → DOCUMENT → APPROVE
  |         |        |          |          |
Identify   Risk    Check     TMP-LOG-005  Operator
equipment  level   types     (program     sign-off
           per     and       document)
           vessel  intervals
```

5.2 Step 1 — Survey the Site

Walk the entire site and identify every pressurised vessel. For each vessel, record:

DATA POINT	SOURCE
Vessel ID (tag number)	Nameplate or P&ID
Description	Visual identification
Location	Physical location on site
Design pressure (PS)	Nameplate
Volume (V)	Nameplate or data sheet
Medium	Operational knowledge
Operating conditions	Normal pressure, temperature, cycles
Safety devices	What protects this vessel
Accessibility	Can it be inspected? Any obstructions?

> If an equipment register already exists (from JDS-PRJ-MEC-001), use it as the starting point. Do not duplicate work.

5.3 Step 2 — Assess Risk Level

For each vessel, determine the supervision intensity:

FACTOR	HIGHER INTENSITY	LOWER INTENSITY
Risk class	Class A	Class B / Simple PV
Medium	Group 1 (dangerous)	Group 2 (non-dangerous)
Operating conditions	Cyclic, high temperature, corrosive	Steady-state, ambient, clean
Age	Old, approaching design life	New, within design life
History	Previous findings, repairs	Clean record
Consequence of failure	Personnel exposure, environmental	Contained, no exposure

The risk assessment drives how often and how thoroughly each vessel is supervised.

5.4 Step 3 — Design Check Types and Intervals

Define what checks to perform and how often. The table below provides baseline intervals — adjust based on risk assessment:

Visual and External Checks

CHECK	BASELINE INTERVAL	WHAT TO LOOK FOR
External surface condition	Monthly	Corrosion, dents, cracks, paint blistering
Insulation condition	Monthly	Damage, moisture ingress, CUI indicators
Support and foundation	Monthly	Settlement, cracking, corrosion of supports
Nameplate legibility	Quarterly	Readable, not obscured
Pressure gauge reading	Daily/shift	Within normal operating range
Temperature reading	Daily/shift	Within design limits
Leak detection	Weekly	Flanges, valves, fittings, weld joints

Safety Device Checks

CHECK	BASELINE INTERVAL	WHAT TO LOOK FOR
Safety valve not gagged or blocked	Weekly	No physical obstruction, no unauthorised isolation
Safety valve seal intact	Monthly	Lead seal or tag present and undisturbed
Safety valve function test	Annually	Lifts at set pressure, reseats cleanly
Rupture disc condition	Monthly	No bulging, corrosion, or damage visible
Pressure switch function	Quarterly	Trips at correct pressure, signal reaches control system

Operational Checks

CHECK	BASELINE INTERVAL	WHAT TO LOOK FOR
Drain operation	Weekly	Condensate drainage, no blockage
Level indicators	Daily/shift	Correct reading, no stuck indication
Vibration / unusual noise	Daily/shift	Change from baseline
Control system alarms	Daily/shift	No standing alarms, correct setpoints

5.5 Step 4 — Document the Program

Use **JDS-TMP-LOG-005** (Supervision Program Template) to create the program document. The template contains all required sections:

1. **Scope** — which equipment is covered
2. **Equipment list** — with risk classification and check assignments
3. **Check schedule** — daily, weekly, monthly, quarterly, annual tasks
4. **Personnel** — who performs what, competence requirements
5. **Documentation** — how rounds are recorded
6. **Findings management** — how deviations are handled
7. **Review schedule** — when the program is reviewed and by whom

5.6 Step 5 — Approve and Issue

The completed supervision program must be:

1. **Reviewed** by a competent person (technical review)
2. **Approved** by the operator (management approval)
3. **Communicated** to all personnel who perform supervision
4. **Filed** in the active programs folder (04-active-programs)
5. **Registered** in JDS-LOG-MEC-005 (Program Register)

6. Performing Supervision Rounds

6.1 Before the Round

1. Review the supervision program for today's scheduled checks
2. Gather equipment: torch, notebook/tablet, PPE, camera
3. Check for any recent findings that need follow-up
4. Prepare a blank round record (JDS-TMP-LOG-006)

6.2 During the Round

1. Follow the check sequence defined in the program
2. For each vessel, work through the assigned checks
3. Record the result of every check: OK, Not OK, or N/A
4. Photograph any findings
5. If a finding is critical (immediate safety concern), stop and escalate

6.3 After the Round

Warning: A **Critical** finding means the vessel must be taken out of service immediately. Notify the operator, isolate the vessel, and raise a corrective action. Do not re-pressurise until resolved.

1. Complete the round record — all fields, sign, and date
2. If findings exist: - **Critical** — vessel out of service, notify operator immediately, raise corrective action - **Non-critical** — log the finding, set a follow-up date, raise corrective action per JDS-PRO-008 - **Observation** — note for monitoring, no immediate action required
3. File the completed record in the client's supervision folder
4. Update the program register (next round date)

6.4 Finding Severity Classification

SEVERITY	DEFINITION	ACTION	TIMELINE
Critical	Immediate risk to personnel or equipment integrity	Take out of service, escalate to operator	Immediate
Major	Condition will deteriorate to critical if not addressed	Plan repair/remediation, monitor closely	Within 30 days
Minor	Condition noted, no immediate risk	Monitor at next round, plan maintenance	Within 90 days
Observation	Not a deficiency, but worth monitoring	Note in record, observe trend	Next round

7. Personnel Competence

7.1 Minimum Requirements

Personnel performing supervision rounds must have:

REQUIREMENT	EVIDENCE
Understanding of pressurised vessel hazards	Training record
Knowledge of the specific equipment supervised	Site familiarisation record
Ability to read and interpret nameplates, gauges, and instruments	Practical assessment
Understanding of the supervision program and check procedures	Program briefing record
Knowledge of emergency procedures	Emergency drill participation

7.2 Competence Records

Competence is managed per **JDS-PRO-009** (Competence Management Procedure). Each person performing supervision must have a current competence record on file.

8. Documentation Requirements

8.1 What Must Be Documented

DOCUMENT	TEMPLATE	RETENTION
Supervision program (the plan)	JDS-TMP-LOG-005	Life of equipment + 5 years
Supervision round records	JDS-TMP-LOG-006	Life of equipment + 5 years
Annual program reviews	JDS-TMP-LOG-007	Life of equipment + 5 years
Corrective action records	JDS-PRO-008	Life of equipment + 5 years
Competence records	JDS-PRO-009	Duration of employment + 5 years
Program register	JDS-LOG-MEC-005	Maintained continuously

8.2 Filing Structure

Each active program in `04-active-programs/` should follow this structure:

```
04-active-programs/
└── [client-name]/
    ├── JDS-LOG-MEC-NNN_supervision-program.md   ← The program document
    ├── rounds/                                   ← Completed round records
    │   ├── YYYY-MM-DD_round-record.md
    │   └── ...
    ├── reviews/                                  ← Annual reviews
    │   ├── YYYY_annual-review.md
    │   └── ...
    └── findings/                                 ← Open and closed findings
        ├── F-001_description.md
        └── ...
```

9. Annual Program Review

9.1 Purpose

The supervision program must be reviewed at least annually to ensure it remains effective and compliant. Use **JDS-TMP-LOG-007** (Annual Review Template).

9.2 Review Checklist

REVIEW ITEM	WHAT TO CHECK
Equipment changes	Any vessels added, removed, or modified?
Operating condition changes	Any changes in pressure, temperature, medium, or duty cycle?
Regulatory changes	Any new amendments or guidance from the authority?
Finding trends	Are the same issues recurring? Does the program catch them?
Personnel changes	Are all supervision personnel still competent and current?
Round completion rate	Were all scheduled rounds performed?
Corrective action closure	Are all findings from the year closed?
Interval adequacy	Are the check intervals appropriate, or should they be adjusted?

9.3 Review Output

The annual review produces: 1. An updated supervision program (new revision) if changes are needed 2. A completed review record (filed in the client's reviews folder) 3. Updated program register entry

10. Integration with JDS-PRJ-MEC-001

This supervision system is designed to work alongside the Pressure Vessel Maintenance Program:

ACTIVITY	SYSTEM
Equipment inventory and classification	JDS-PRJ-MEC-001
Formal inspection planning and scheduling	JDS-PRJ-MEC-001
Ongoing supervision between inspections	**JDS-PRJ-MEC-002 (this project)**
Inspection report management	JDS-PRJ-MEC-001
Corrective actions	JDS-PRO-008 (shared)
Competence management	JDS-PRO-009 (shared)

11. References

DOCUMENT	PURPOSE
JDS-RPT-MEC-003	AFS 2017:3 Consolidated Summary
JDS-PRO-008	Corrective Action Procedure
JDS-PRO-009	Competence Management Procedure
JDS-PRO-010	Ongoing Maintenance Program Procedure
JDS-PRJ-MEC-001	Pressure Vessel Maintenance Program
JDS-TMP-LOG-005	Supervision Program Template
JDS-TMP-LOG-006	Supervision Round Record Template
JDS-TMP-LOG-007	Annual Review Template

Revision History

REV	DATE	AUTHOR	DESCRIPTION
A	2026-04-10	N. Johansson	Initial release — complete supervision program methodology